Identities, Multiple Angles and Exact Solution Sets Answer Key

Precalculus

Quick Answers

1. $\sin(\theta)$
2. $\frac{\sqrt{6}+\sqrt{2}}{4}$
3. $\frac{24}{25}$
4. 30, 330 degrees
5. $2-\sqrt{3}$
6. 0, 60, 180, 300 degrees
7. $\frac{2}{\sin^2(\theta)}$
8. 60, 180, 300 degrees
9. 30, 90, 150, 270 degrees
10. $\frac{\sqrt{2}-\sqrt{6}}{4}$
11. (a) simplified single-term form = $\frac{2}{\sin(\theta)}$, —
12. (a) factored form = $(1-2\sin(\theta))(\sin(\theta)-1)$, (b) exact solution set = 30, 90, 150 degrees, —

What is verified

13 of 15 answers machine-checked against SymPy, across 12 problems.
— marks an answer only you can judge — problems 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus
HSF-TF.C.8 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 15 tagged checks.

1. Simplify $\dfrac{1-\cos^2\theta}{\sin\theta}$.

Solution: the numerator is the Pythagorean identity rearranged: $1-\cos^2\theta = \sin^2\theta$. Dividing, $\dfrac{\sin^2\theta}{\sin\theta} = \sin\theta$ wherever $\sin\theta \neq 0$. $\boxed{\sin\theta}$

2. Exact value of $\cos 15°$.

Solution: write $15° = 45° - 30°$ and use $\cos(A-B) = \cos A\cos B + \sin A \sin B$:

$$\cos 15° = \cos 45° \cos 30° + \sin 45° \sin 30°$$
$$= \frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\cdot\frac{1}{2} = \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4}.$$

$$\boxed{\frac{\sqrt{6}+\sqrt{2}}{4}}$$

3. $\sin\theta = \dfrac{3}{5}$, θ in quadrant I. Find $\sin 2\theta$.

Solution: first get the cosine from the Pythagorean identity, taking the positive root because quadrant I has cosine positive: $\cos\theta = \sqrt{1-\left(\frac{3}{5}\right)^2} = \sqrt{\frac{16}{25}} = \frac{4}{5}$. Then

$$\sin 2\theta = 2\sin\theta\cos\theta = 2\cdot\frac{3}{5}\cdot\frac{4}{5}.$$

$$\boxed{\frac{24}{25}}$$

4. Solve $2\cos\theta - \sqrt{3} = 0$ on $0^\circ \le \theta < 360^\circ$.

Solution: $\cos\theta = \dfrac{\sqrt{3}}{2}$, reference angle 30°, and cosine is positive in quadrants I and IV, so $\theta = 30^\circ$ and $\theta = 360^\circ - 30^\circ = 330^\circ$. $\boxed{\theta = 30^\circ,\ 330^\circ}$

5. Exact value of $\tan 15^\circ$.

Solution: with $\tan(A-B) = \dfrac{\tan A - \tan B}{1+\tan A\tan B}$ and $15^\circ = 45^\circ - 30^\circ$:

$$\tan 15^\circ = \frac{1-\frac{1}{\sqrt{3}}}{1+\frac{1}{\sqrt{3}}} = \frac{\sqrt{3}-1}{\sqrt{3}+1}.$$

Multiply above and below by $\sqrt{3}-1$: the numerator becomes $(\sqrt{3}-1)^2 = 4-2\sqrt{3}$ and the denominator becomes $3-1=2$. $\boxed{2-\sqrt{3}}$

6. Solve $\sin 2\theta = \sin\theta$ on $0^\circ \le \theta < 360^\circ$.

Solution: expand the double angle, collect on one side, and factor — never divide by $\sin\theta$.

$$2\sin\theta\cos\theta - \sin\theta = 0 \quad\Longrightarrow\quad \sin\theta\,(2\cos\theta - 1) = 0$$

$\sin\theta = 0$ gives $0^\circ, 180^\circ$; $\cos\theta = \dfrac{1}{2}$ gives $60^\circ, 300^\circ$. $\boxed{\theta = 0^\circ,\ 60^\circ,\ 180^\circ,\ 300^\circ}$

7. Simplify $\dfrac{1}{1+\cos\theta} + \dfrac{1}{1-\cos\theta}$.

Solution: the common denominator is a difference of squares, which is exactly where the Pythagorean identity enters:

$$\frac{(1-\cos\theta)+(1+\cos\theta)}{(1+\cos\theta)(1-\cos\theta)} = \frac{2}{1-\cos^2\theta} = \frac{2}{\sin^2\theta}.$$

$$\boxed{\frac{2}{\sin^2\theta}}$$

8. Solve $2\cos^2\theta + \cos\theta - 1 = 0$ on $0^\circ \le \theta < 360^\circ$.
Solution: this is a quadratic in $\cos\theta$; factor it as one.

$$(2\cos\theta - 1)(\cos\theta + 1) = 0$$

$\cos\theta = \dfrac{1}{2}$ gives $60^\circ, 300^\circ$; $\cos\theta = -1$ gives 180° — once only, because -1 is the minimum of cosine. $\boxed{\theta = 60^\circ,\ 180^\circ,\ 300^\circ}$

9. Solve $\sin 2\theta = \cos\theta$ on $0^\circ \le \theta < 360^\circ$.
Solution: expand and factor, keeping the common $\cos\theta$ as a factor.

$$2\sin\theta\cos\theta - \cos\theta = 0 \quad\Longrightarrow\quad \cos\theta\,(2\sin\theta - 1) = 0$$

$\cos\theta = 0$ gives $90^\circ, 270^\circ$; $\sin\theta = \dfrac{1}{2}$ gives $30^\circ, 150^\circ$. $\boxed{\theta = 30^\circ,\ 90^\circ,\ 150^\circ,\ 270^\circ}$

10. Exact value of $\cos 105^\circ$.
Solution: $105^\circ = 60^\circ + 45^\circ$, and $\cos(A+B) = \cos A\cos B - \sin A\sin B$:

$$\begin{aligned}\cos 105^\circ &= \cos 60^\circ\cos 45^\circ - \sin 60^\circ \sin 45^\circ\\ &= \frac{1}{2}\cdot\frac{\sqrt{2}}{2} - \frac{\sqrt{3}}{2}\cdot\frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4}.\end{aligned}$$

The result is negative, as it must be for a quadrant II angle. $\boxed{\dfrac{\sqrt{2}-\sqrt{6}}{4}}$

11. Identity: $\dfrac{\sin\theta}{1+\cos\theta} + \dfrac{1+\cos\theta}{\sin\theta} = \dfrac{2}{\sin\theta}$. (a) Simplify the left side. (b) Prove it.
(a) Over the common denominator $\sin\theta\,(1+\cos\theta)$:

$$\frac{\sin^2\theta + (1+\cos\theta)^2}{\sin\theta\,(1+\cos\theta)} = \frac{\sin^2\theta + 1 + 2\cos\theta + \cos^2\theta}{\sin\theta\,(1+\cos\theta)} = \frac{2+2\cos\theta}{\sin\theta\,(1+\cos\theta)}.$$

Factor the numerator as $2(1+\cos\theta)$ and cancel. $\boxed{\dfrac{2}{\sin\theta}}$

(b) *Instructor-judged.* A correct proof works on the left-hand side only and names each move: common denominator $\sin\theta(1+\cos\theta)$; expand $(1+\cos\theta)^2$; apply $\sin^2\theta+\cos^2\theta = 1$ to collapse the numerator to $2+2\cos\theta$; factor out 2 and cancel the shared $(1+\cos\theta)$; state that the identity holds wherever $\sin\theta \neq 0$ and $\cos\theta \neq -1$. A chain of equalities that manipulates both sides at once, or a check at one angle, is not a proof.

12. $\cos 2\theta + 3\sin\theta = 2$ on $0^\circ \le \theta < 360^\circ$. (a) Rewrite and factor. (b) Solve. (c) Explain the count.

(a) Use the form of the double angle that produces sine only, $\cos 2\theta = 1 - 2\sin^2\theta$:

$$1 - 2\sin^2\theta + 3\sin\theta - 2 = -2\sin^2\theta + 3\sin\theta - 1 = (1 - 2\sin\theta)(\sin\theta - 1).$$

$$\boxed{(1 - 2\sin\theta)(\sin\theta - 1)}$$

(b) $1 - 2\sin\theta = 0$ gives $\sin\theta = \dfrac{1}{2}$, so $\theta = 30^\circ$ and 150°; $\sin\theta - 1 = 0$ gives $\theta = 90^\circ$.

$$\boxed{\theta = 30^\circ,\ 90^\circ,\ 150^\circ}$$

(c) *Instructor-judged.* A correct response explains the count factor by factor: $\sin\theta = \dfrac{1}{2}$ is met twice in a full turn (quadrants I and II), while $\sin\theta = 1$ is the maximum of sine and is met exactly once, at 90° — two plus one is three. Saying only that the quadratic has two roots, without noting that one root yields a single angle, is partial credit.